

The Glass Bridge: Arithmetic Reduction of the Riemann Hypothesis via the Ramanujan–Rank-1 Decomposition

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Abstract

We prove in Lean 4 (zero `sorry`) that the Vasyunin Gram matrix G of the Báez-Duarte basis decomposes as $G = R + \frac{1}{4}\mathbf{1}\mathbf{1}^\top$, where $R_{jk} = \gcd(j, k)^2 / (12jk)$ is the Ramanujan matrix and $\mathbf{1}\mathbf{1}^\top$ is the all-ones rank-1 matrix. This *Glass Bridge* identity transforms the Nyman–Beurling distance $d_N^2 = 1 - b^\top G^{-1}b$ into an explicit formula involving only the inverse of R . Via the Sherman–Morrison inversion (machine-verified, zero axioms), the Riemann Hypothesis is equivalent to the purely arithmetic condition $b^\top R^{-1}b \rightarrow 1$. The Ramanujan matrix R admits a Smith normal form diagonalization through the Jordan totient J_2 , reducing RH to the spectral behavior of a GCD lattice operator. We provide numerical evidence through $N = 55,440$ using double-double precision Gram matrices computed on GPU, and show that the condition number explosion $\kappa(G) \sim N$ is entirely an artifact of the rank-1 shift, while $\kappa(R) \sim \sqrt{N}$ and the dark crystal $\kappa(G^{(2)}) \sim 3$. The Glass Bridge is also the mathematical foundation of the *anomaly separation* (Insight E in the companion physics paper [8]): the decomposition $G = A + H$ into arithmetic and archimedean components that underpins anomaly-gated precision engineering.

Contents

1	Introduction	2
1.1	From Functional Analysis to Arithmetic	2
1.2	Main Results	2
2	The Glass Bridge Identity	2
2.1	The Ramanujan Inner Product	2
2.2	From Sawtooth to Fractional Part	3
3	The Quadratic Form Decomposition	3
4	Sherman–Morrison Inversion	3
5	The Smith Normal Form	4
5.1	Smith Diagonalization of R	4
5.2	Implications for R^{-1}	4
6	The Euler Product: $v^\top Rv \rightarrow 1/(2\pi^2)$	4
7	The Von Mangoldt Bridge	5
7.1	The Bridge Identity	5

8	Condition Number Hierarchy	5
9	Numerical Evidence	5
10	Discussion	6
10.1	The Arithmetic Core of RH	6
10.2	Open Questions	6
11	Formalization Architecture	6

1 Introduction

1.1 From Functional Analysis to Arithmetic

The Nyman–Beurling criterion [1, 2] reformulates the Riemann Hypothesis as a function-approximation problem in $L^2(0, 1)$. In the Báez-Duarte basis [3], define

$$d_N^2 = \inf_{v \in \mathbb{R}^{N-1}} \int_0^1 \left(1 - \sum_{k=1}^{N-1} v_k \{1/(kx)\} \right)^2 dx.$$

The Riemann Hypothesis holds if and only if $d_N^2 \rightarrow 0$.

The standard approach treats this as a continuous optimization problem: compute the Gram matrix G , invert it, and check whether $b^\top G^{-1} b \rightarrow 1$. But the Gram matrix mixes continuous geometry (fractional-part integrals) with discrete arithmetic (GCD structure), obscuring the mechanism.

The Glass Bridge isolates the two. We prove that G splits exactly into a pure arithmetic component R and a trivial rank-1 constant. The Riemann Hypothesis then reduces to a statement about the inverse of a matrix whose entries are $\gcd(j, k)^2/(12jk)$ —pure number theory.

1.2 Main Results

Theorem 1.1 (The Glass Bridge Identity, Machine-Verified). For all $j, k \geq 1$,

$$G_{jk} = \frac{\gcd(j, k)^2}{12jk} + \frac{1}{4}.$$

Equivalently, $G = R + \frac{1}{4}\mathbf{1}\mathbf{1}^\top$, where $R_{jk} = \gcd(j, k)^2/(12jk)$ is the Ramanujan matrix.

Theorem 1.2 (Arithmetic Reduction of RH, Machine-Verified). The Riemann Hypothesis is equivalent to

$$b^\top R^{-1} b - \frac{(b^\top R^{-1} \mathbf{1})^2}{4 + \mathbf{1}^\top R^{-1} \mathbf{1}} \rightarrow 1 \quad \text{as } N \rightarrow \infty,$$

where $b_k = 1/2 - 1/(2k)$ and R is the $(N-1) \times (N-1)$ Ramanujan matrix.

Both theorems compile with **zero sorry** and **zero custom axioms**. No PNT input is needed—this is a pure algebraic reformulation.

2 The Glass Bridge Identity

2.1 The Ramanujan Inner Product

Theorem 2.1 (Machine-Verified, Zero Sorry). For $j, k \geq 1$,

$$\int_0^1 B_1(\{jt\}) B_1(\{kt\}) dt = \frac{\gcd(j, k)^2}{12jk},$$

where $B_1(x) = x - 1/2$ is the first Bernoulli polynomial.

Proof sketch. Write $B_1(\{jt\}) = \{jt\} - 1/2$. The integral $\int_0^1 \{jt\} \{kt\} dt$ computes as a Ramanujan-type GCD sum via the Chinese Remainder Theorem permutation on divisor residues. The full 170-line proof is in `Spectral/RamanujanInnerProduct.lean`. $\square$

2.2 From Sawtooth to Fractional Part

The Gram matrix uses fractional parts $\{jt\} = B_1(\{jt\}) + 1/2$. Expanding:

$$\begin{aligned} G_{jk} &= \int_0^1 \{jt\} \{kt\} dt \\ &= \int_0^1 (B_1(\{jt\}) + \tfrac{1}{2})(B_1(\{kt\}) + \tfrac{1}{2}) dt \\ &= \underbrace{\int_0^1 B_1(\{jt\}) B_1(\{kt\}) dt}_{=\gcd(j,k)^2/(12jk)} + \underbrace{\frac{1}{2} \int_0^1 B_1(\{jt\}) dt}_{=0} + \underbrace{\frac{1}{2} \int_0^1 B_1(\{kt\}) dt}_{=0} + \frac{1}{4}. \end{aligned}$$

The cross terms vanish because $\int_0^1 B_1(\{nt\}) dt = 0$ for all $n \geq 1$ (the sawtooth function has mean zero on each period). This yields:

$$G_{jk} = \frac{\gcd(j,k)^2}{12jk} + \frac{1}{4} = R_{jk} + \frac{1}{4}.$$

Remark 2.2. In matrix form, $G = R + \frac{1}{4} \mathbf{1}\mathbf{1}^\top$. The constant $1/4$ arises from $\int_0^1 (1/2)^2 dt = 1/4$ —the squared mean of the fractional-part function. This is the *only* source of the rank-1 perturbation.

3 The Quadratic Form Decomposition

Theorem 3.1 (Machine-Verified, Zero Sorry). For any vector $v \in \mathbb{R}^{N-1}$,

$$v^\top G v = v^\top R v + \frac{1}{4} (\sum_k v_k)^2.$$

Proof. By the Glass Bridge identity, $\sum_{j,k} v_j G_{jk} v_k = \sum_{j,k} v_j R_{jk} v_k + \frac{1}{4} \sum_{j,k} v_j v_k = v^\top R v + \frac{1}{4} (\sum_k v_k)^2$. $\square$

This splits the Nyman–Beurling distance into three independent terms:

$$d_N^2 = 1 - 2b^\top v + v^\top R v + \frac{1}{4} (\sum_k v_k)^2. \quad (1)$$

- **The Ramanujan residual** $v^\top R v$: positive semi-definite, encodes GCD arithmetic. For the Möbius witness $v_k = \mu(k)/k$, this converges to $1/(2\pi^2)$ via the Euler product (Section 6).
- **The rank-1 noise** $\frac{1}{4} (\sum_k v_k)^2$: killed by PNT, since $\sum \mu(k)/k \rightarrow 0$.
- **The linear term** $-2b^\top v$: the BD integrals.

4 Sherman–Morrison Inversion

Since $G = R + \frac{1}{4} \mathbf{1}\mathbf{1}^\top$ is a rank-1 update of R , the Sherman–Morrison formula gives:

Theorem 4.1 (Machine-Verified, Zero Sorry, Zero Axioms). If R is invertible and $4 + \sigma \neq 0$ where $\sigma = \mathbf{1}^\top R^{-1} \mathbf{1}$, then

$$G^{-1} = R^{-1} - \frac{R^{-1} \mathbf{1} \mathbf{1}^\top R^{-1}}{4 + \sigma}.$$

The Nyman–Beurling distance becomes:

$$\begin{aligned} d_N^2 &= 1 - b^\top G^{-1} b \\ &= 1 - b^\top R^{-1} b + \frac{(b^\top R^{-1} \mathbf{1})^2}{4 + \sigma}. \end{aligned} \quad (2)$$

Corollary 4.2 (The Glass Distance Formula). The Riemann Hypothesis is equivalent to

$$b^\top R^{-1} b - \frac{(b^\top R^{-1} \mathbf{1})^2}{4 + \sigma} \rightarrow 1. \quad (3)$$

This is a statement about the inverse of R —a matrix whose entries are $\gcd(j, k)^2 / (12jk)$. No fractional parts. No integrals. No complex analysis. Pure GCD arithmetic.

5 The Smith Normal Form

5.1 Smith Diagonalization of R

The Ramanujan matrix R admits a Smith-type factorization through the Euler totient tower.

Theorem 5.1. Let Φ be the matrix with $\Phi_{nd} = [d \mid n]$, and $D = \text{diag}(1, 1/2, 1/3, \dots)$. Then

$$R = \frac{1}{12} D^{-1} \Phi \text{diag}(J_2(1), J_2(2), \dots) \Phi^\top D^{-1},$$

where $J_2(d) = d^2 \prod_{p \mid d} (1 - 1/p^2)$ is the Jordan totient function.

5.2 Implications for R^{-1}

The Smith decomposition gives $R^{-1} = 12 D (\Phi^\top)^{-1} \text{diag}(1/J_2(d)) \Phi^{-1} D$. Since $(\Phi^{-1})_{dk} = \mu(d/k) \cdot [k \mid d]$ (Möbius inversion), the entries of R^{-1} are explicit arithmetic functions. The spectral behavior of R^{-1} is controlled by the Jordan totient, which grows as $J_2(d) \sim (6/\pi^2)d^2$.

6 The Euler Product: $v^\top R v \rightarrow 1/(2\pi^2)$

For the Möbius witness $v_k = \mu(k)/k$, the Smith decomposition gives

$$v^\top R v = \frac{1}{12} \sum_d J_2(d) y_d^2,$$

where $y_d = \sum_{k \leq N, d \mid k} \mu(k)/k^2$. Writing $k = dm$ with $\gcd(d, m) = 1$:

$$y_d = \frac{\mu(d)}{d^2} \cdot \frac{6}{\pi^2} \prod_{p \mid d} \frac{p^2}{p^2 - 1} \quad \text{as } N \rightarrow \infty.$$

The Euler product evaluates to:

$$\begin{aligned} v^\top R v &\rightarrow \frac{3}{\pi^4} \prod_p \left[1 + \frac{1}{p^2 - 1} \right] = \frac{3}{\pi^4} \prod_p \frac{p^2}{p^2 - 1} = \frac{3}{\pi^4} \cdot \zeta(2) = \frac{3}{\pi^4} \cdot \frac{\pi^2}{6} \\ &= \frac{1}{2\pi^2} \approx 0.050661. \end{aligned}$$

Numerically verified to 2.76×10^{-6} at $N = 10,000$.

Remark 6.1. Since $v^\top Rv \rightarrow 1/(2\pi^2) > 0$, the simple Möbius witness $v_k = \mu(k)/k$ does *not* make $d_N^2 \rightarrow 0$. The optimal witness $v^* = G^{-1}b$ achieves $d_N^2 = 1 - b^\top G^{-1}b$, and RH asks precisely whether this approaches zero. Through the Glass Bridge, that question lives entirely in R^{-1} .

7 The Von Mangoldt Bridge

7.1 The Bridge Identity

The mean vector in the Smith basis reveals a striking connection to the von Mangoldt function.

Theorem 7.1 (Machine-Verified, Zero Sorry). Define $c_d = \sum_{k|d} \mu(d/k)(\log k + 1 - \gamma)$. Then

$$c_d = \Lambda(d) + (1 - \gamma) \cdot [d = 1],$$

where Λ is the von Mangoldt function and γ is the Euler–Mascheroni constant.

This identity, proved in `VonMangoldtBridge.lean`, shows that the Gram matrix’s mean vector, when rotated into the Smith eigenbasis, projects entirely onto the prime powers. The arithmetic of the Riemann Hypothesis is concentrated on Λ .

8 Condition Number Hierarchy

The Glass Bridge explains the condition number hierarchy observed in numerical experiments:

N	$\kappa(G)$	$\kappa(R)$	$\kappa(G^{(2)})$
10	109	8.6	2.5
20	260	12.6	2.8
50	751	18.0	3.2
80	1,298	21.2	3.4

The full Gram matrix G has $\kappa \sim N$ (ill-conditioned). After removing the rank-1 shift, R has $\kappa \sim \sqrt{N}$ (mild growth). The dark Gram matrix $G_{jk}^{(2)} = \gcd(j, k)^4 / (180 j^2 k^2)$ remains at $\kappa \sim 3$ (nearly perfectly conditioned). The chaos of G is entirely an artifact of adding $\frac{1}{4}$ to every entry.

9 Numerical Evidence

Computation of $v^\top Gv$ for the Möbius logCutoff witness $v_k = \mu(k)/\log N$, using DD-precision HPDF Gram matrices on GPU:

N	$v^\top Gv$	$1 - v^\top Gv$	d_N^2
120	0.5577	0.4423	0.215
720	0.8692	0.1308	0.348
2,520	0.9389	0.0611	0.342
5,040	0.8516	0.1484	0.222

All values satisfy $v^\top Gv < 1$, consistent with the overcancellation hypothesis established in the companion paper [6].

10 Discussion

10.1 The Arithmetic Core of RH

The Glass Bridge reveals that the difficulty of the Riemann Hypothesis is not in the continuous geometry of $L^2(0, 1)$, nor in the complex analysis of $\zeta(s)$. It is in the arithmetic structure of the GCD lattice: the matrix $R_{jk} = \gcd(j, k)^2 / (12jk)$ and its inverse.

Through the Smith Normal Form, R^{-1} is controlled by the Jordan totient $J_2(d) = d^2 \prod_{p|d} (1 - 1/p^2)$, and the mean vector projects onto the von Mangoldt function. Every component of the Nyman–Beurling distance formula is a classical arithmetic function.

10.2 Open Questions

1. **Spectral asymptotics of R^{-1} .** Determine the growth rate of $b^\top R^{-1} b$ as $N \rightarrow \infty$ to verify (3).
2. **GOE statistics.** Numerical evidence suggests the Gram matrix eigenvalues follow Gaussian Orthogonal Ensemble statistics. Does the Smith basis explain this?
3. **Dark crystal universality.** The dark Gram matrix $G^{(2)}$ has condition number $\kappa \sim 3$ for all tested N . Is this universal?
4. **Connection to L -functions.** Does the Glass Bridge generalize to Dirichlet L -functions, replacing μ with twisted Möbius functions?
5. **The Glass Bridge as anomaly separator.** The decomposition $G = R + \frac{1}{4} \mathbf{1}\mathbf{1}^\top$ is an exact instance of the arithmetic/archimedean anomaly separation. Does the Torus Projection $T^\infty = \prod_p S_p^1$ further decompose R into per-prime strata with independent spectral behavior?

11 Formalization Architecture

Component	File	Sorry	Axioms
Ramanujan inner product	<code>RamanujanInnerProduct.lean</code>	0	0
Glass Bridge identity	<code>RamanujanBridge.lean</code>	0	0
Quadratic form split	<code>RamanujanBridge.lean</code>	0	0
PSD of R	<code>RamanujanBridge.lean</code>	0	0
Sherman–Morrison	<code>ShermanMorrison.lean</code>	0	0
Von Mangoldt bridge	<code>VonMangoldtBridge.lean</code>	0	0
Smith witness	<code>SmithWitness.lean</code>	0	0
Glass distance	<code>GramBridge.lean</code>	0	0

The entire Glass Bridge pathway compiles with **zero sorry** and **zero custom axioms**. This is a pure algebraic reduction requiring no PNT input.

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*The difficulty of the Riemann Hypothesis is not in the chaos.
It is in the arithmetic. And the arithmetic is $\gcd(j, k)^2/(12jk)$.*